First: Short Framing on Climate Change

Please read the following text carefully. The technology explained on the next page refers to it:

According to the latest report by the Intergovernmental Panel on Climate Change (IPCC), the global average surface temperature has risen at a rate of 1°C (1.8°F) compared to 1900. It is extremely likely that this effect is caused by increased emissions of greenhouse gases such as carbon dioxide (CO_2). Greenhouse gas emissions are released for example through the burning of coal, oil and gas. One degree may not seem like much, but it already has an impact, which can worsen in the future as the temperature continues to rise:

- Increasingly extreme weather events, like heavy rainfall and storms.
- The oceans are warming due to human influence, which is leading to the melting of the polar ice caps.
- Increased acidity of the ocean, because the water takes up too much of the released CO_2, which is leading to the destruction of corals reefs worldwide.

If greenhouse gas emissions continue as they are right now, by 2100 the global temperature could rise up to 3.3 – 5.7°C (5.94 – 10.26°F) compared to before 1900.

Second: non-fictional technology: 'Stratospheric Aerosol Injection'

Please read the following text carefully. The questions later in the study refer to it.

Imagine you find yourself in the year 2030. Despite worrying reports, greenhouse gases have not been sufficiently reduced and governments around the world are beginning to use new technologies to halt global warming:

The scenario describes a typical day in the life of Charlie, a citizen facing new technological developments to stop further global warming. Charlie is concerned about the technology, but also sees hope that climate change could be stopped.

On a Saturday in the year 2030, Charlie looks up into the sky and sees planes releasing sulfur particles in higher regions of the atmosphere at around 50.000 feet (15 kilometers). Charlie remembers that this sulfur particles reflect sunlight partially back out into space before it warms the earth. He knows that this technique is called "Stratospheric Aerosol Injection", which aims to reflect incoming heat from the sun brought by sunlight back into space to prevent further warming of land and oceans. The cooling effect of this technique was demonstrated by past volcanic eruptions, because when a volcano erupts, sulfur is also released into the atmosphere. To the best of Charlie's knowledge sulfur is a chemical element found in almost all life forms, and is what gives eggs their distinctive smell for example. Charlie looks at the planes in the sky and thinks about the advantages of "Stratospheric Aerosol Injection" (SAI):

- SAI is feasible and potentially very effective
- compared to other technologies the costs of SAI are low
- when SAI is deployed, it should lower temperatures within a year

However, Charlie is very concerned that this technique also comes with certain risks and uncertainties:

- SAI needs to be deployed for decades or centuries – otherwise temperature could rise even more dramatically within just a few years
- SAI does nothing to counter the inherent cause of climate change – the concentration of carbon dioxide (CO_2) in the atmosphere - and therefore it does not reduce the acidity of the ocean

- There are already political and social conflicts over its use, as single countries deploying SAI but affecting the whole Earth

Charlie feels frightened, because potential side effects have not yet been fully resolved and it is unclear what will happen in the future. This technique is used as an emergency solution, because the temperature has risen beyond the 2°C (3.6°F) target, which nearly all countries in the world have agreed upon. Charlie reflects that, despite SAI we have to change our lifestyles and transform the economy in a way that limits global warming even further.